

Proficient Self-Adaptive Dynamic Traffic Monitoring Control System

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Abstract – The conventional traffic light control frameworks are governed by fixed time interims of the traffic lights. These traditional fixed traffic light controllers have certain constraints and are less effective on the grounds that they utilize an equipment, which have capacities as indicated by the program that comes up with limitations of the inconsistent adaptability of adjustments. Along these lines, due to the fixed time interims of green and red signs, there is overabundance average waiting time and superfluous holding up time on lanes, due to which vehicles tend to devour more fuel. This in the long run indicates the ecological contamination and makes a few medical related problems among the individuals on street and living close by, as they tend to inhale the impure air. Likewise, these customary traffic light control frameworks do not have any advancements or arrangements to give any data on traffic densities on different streets, which prompt traffic clogs. Consequently, to make the traffic light scheme control and traffic guideline progressively productive, the proposed work makes use of the development of a novel strategy called as Adaptive Traffic Light Control System, modelled with highly proficient framework. The proposed framework utilizes a complete system of cluster of sensors for detecting the traffic with priority based traffic management. On positioning this effectively detected traffic, the planning interims of red and green light signals across the streets are cleverly chosen and changed in order to keep the holding up time least. In this way, enhancement of the traffic light with periodically exchanges, builds the street limit, spares time for voyaging and forestalls traffic blockages. The traffic related details are intermittently updated to the multi-cloud environment. The model is assessed by means of rebuilding in the Simulation of Urban Mobility framework (SUMO) in a vehicular system using Reinforced Decision Process (RDP), and the reenactment results show the significantly improved efficiency of the model in controlling and regulating the traffic with dynamic and self-adaptive management.

Keywords - Traffic Monitoring, Reinforcement Learning, Multi-Cloud Environment, Edge detection, Urban Mobility, Image Processing.

I. INTRODUCTION

Efficient traffic management system for urban mobility is the need of the hour. Conventional traffic system delays the regular traffic and increases the average waiting time of vehicles in succession. There is no priority based traffic diversion scheme in conventional systems and some vehicles tend to disobey the traffic signals and surpass the waiting vehicles. The waiting time and throughput of vehicles in intersection of lanes in case of conventional traffic management system is quite unpredictable [1]. This leads to traffic congestion, deadlocks and prolonged waiting time of vehicles on the lane. The junction of intersection becomes a havoc with traffic violation and increasing traffic densities [2].

The conventional fixed time intervals of traffic lights contribute to increased average waiting time of vehicles in lane and intersections. This also tends to proportionally increase the fuel consumption of waiting vehicles and leads to increase in CO₂ emission in the atmosphere, which when inhaled leads to chronic problems in humans [3]. This necessitates the need to develop a highly efficient framework for self-adaptive and dynamic traffic monitoring and control system, which can provide a one-stop solution to all traffic related problems. The main concern of vehicles waiting in the lane is the longer average waiting time of vehicles in queue with more “traffic build-ups”. The other concern is with respect to the enormously increasing fuel consumption and the subsequent fuel cost incurred to the vehicle. The conventional traffic regulation systems face these challenges and are more prone to increased waiting time and fuel cost [4].

In high traffic situations, a human administrator monitors the constant traffic condition in the intersection or junction and intelligently decides the length of the permitted sitting back for every course utilizing his/her long haul understanding and comprehension about the mid-crossing point, which is exceptionally viable [5]. This perception spurs us to propose a savvy convergence traffic light administration framework which can take continuous traffic condition as info and figure out how to deal with the crossing point simply like the human administrator. To execute such a framework, the observer requires the 'eyes' to watch the continuous street condition and 'a cerebrum' to process it. For the former, late advances in sensors and systems administration innovation empowers taking continuous traffic data as information, for example, the quantity of vehicles, the areas of vehicles, and their holding up time are to be monitored [5].

The Average Waiting Time (AWT) of vehicles is based on the amount of time spent by the vehicle without movement in the traffic, until and unless the traffic signal goes on. The more the AWT, the more is the travel time as well as the fuel cost. The prevailing systems do not consider these factors and hence are not contributing to the efficiency of traffic management framework. Apart from these factors, the system modelled tends to consider and address the issue of priority based signalling to give utmost preference and priority to emergency vehicles to head first, especially in case of urban mobility.

The paper is organized into various sections as follows. Section I gives an Introduction about the need for traffic management systems. Section II deals with the explanation of the various existing models. Section III enumerates the proposed method with layout architecture. Section IV gives the

experimental results. Section V summarizes the contribution of the paper with the conclusion and future work.

II. RELATED WORKS

This section deals with the review of related works. The prevailing traffic light control either sends fixed programs without considering ongoing traffic or considering the traffic to a restricted degree. In [5], the fixed programs set the traffic signals equivalent time length in each cycle, or distinctive time span dependent on chronicled data. Some control programs take contributions from sensors, for example, underground inductive circle finders to identify the presence of vehicles before traffic lights [6]. Be that as it may, the sources of information are handled in an extremely coarse manner to decide the span of green/red lights. At times, existing traffic light control frameworks work, however at a low efficiency [7]. With respect to this and in numerous different cases, for example, a football occasion or an increasingly basic high traffic hour scenario, the traffic light control frameworks become deadened. Rather, the conventional system regularly witnesses a police officer legitimately dealing with the convergence by waving signals [8]. The Average Waiting Time (AWT) of vehicles in these conventional systems especially in the intersection of lanes is highly unpredictable. The existing schemes are not much adaptive to traffic flow and have to be continuously monitored.

Existing inefficient traffic light control systems face certain challenges and constraints, like extended traffic queue length with long deferral and exploitation of vitality. To improve efficiency, taking ongoing traffic data as an information and powerfully altering the traffic light span as needs be is an absolute necessity [9]. As far as how to robustly modify traffic signal's length, existing works either split the traffic signal into equivalent term or concentrate restricted traffic data from the genuine information. The best way is to have a clear understanding on how the system can select the traffic signal's duration, which actually depends on the data gathered from various sensors and vehicular networks [10].

Conventional models have wasteful traffic light control which causes various issues like long postponement and misuse of vitality [11]. To improve the proficiency, there is an absolute necessity to take the constant traffic data as an input information and robustly amending the traffic light accordingly. The existing works fail to address on how to dynamically transform traffic signs' span, existing works either split the traffic signal into equivalent term or concentrate restricted traffic data from the genuine information [12]. The current framework chooses the traffic signs' length, which is dependent on the gathered information from various sensors and vehicular systems. In [13], an intelligent traffic management system model is developed, which attempts to decrease the Average Waiting Time (AWT) of vehicles, but the prioritization among the vehicles based on emergency conditions is not addressed in this work. The dynamic traffic update is another factor that needs to be considered in case of the prevailing systems, to properly adhere the rules and regulate the traffic with smooth functioning of the transport system [14].

III. PROPOSED SYSTEM

The proposed system utilizes a deep reinforcement learning model represented by the Reinforced Decision Process (RDP) algorithm, which enables the proper monitoring and control of the traffic lights. In the model, the traffic situation which is complex in nature is presented as various states of the process control is evaluated, by collecting the required information and dynamically partitioning the entire convergence into smaller networks [15]. The changes related to the activities of planning of traffic light scenarios, as a means of high-priority measurement of Markov decision based procedure. The prize is the aggregate holding up time distinction between two cycles. To illuminate the model, a conventional neural system is utilized to outline states to rewards. The system uses image processing to control traffic. The density of traffic in lanes is a factor to be considered and measured for monitoring the traffic and regulating the same without causing havoc. The traffic density is measured by detecting the images of the lane using camera and the image captured is compared with the reference sample images of lanes captured without any traffic. This model is simulated in such an environment to discover the traffic densities on roads to allot the time period of green signal more intelligently for each lane.

The versatile idea of this framework ends up being exceptionally proficient in managing the variable rush hour gridlock densities on streets. Additionally, by giving traffic thickness data to drivers the traffic can be very much overseen. A directed traffic stays away from blockages and forestalls mishaps.

The model makes use of the image processing simulation environment for calculation of traffic density as cameras are readily available infrastructure on road intersections.

If there is an empty lane, then the model generates the green traffic light signal, in order to facilitate passer-by vehicles to cross the lane without any hindrance and abated waiting time.

If the lane has many vehicles lined-up on the road, then the red traffic light signal is enabled for prioritizing the movement of vehicles and properly regulating the traffic in the lane. This in turn increases the throughput and overall performance of the system by properly channelizing the traffic.

Emergency vehicles like ambulance are given the highest priority with the image of the vehicle captured and processed accordingly. Sensor based traffic management with effective image capture of emergency vehicles for priority based signaling is considered. This will save precious lives and property by giving way and preference to emergency services like ambulance, fire and police. This will not cause any chaos and emergency services can be delivered smoothly and without any delay.

The work proposes a profound fortification learning model to control the traffic light. In this model, dimensions of the intricate traffic situation, taken as states is implemented by conglomerating the information and separating the complete convergence into smaller shards of frameworks. The designing of traffic light changes are the activities, which are presented as large-scale high priority Markov decision choice procedure.

The proposed system is enhanced with additional parameters for performance improvement namely, prioritizing traffic flow with two-fold based Q-learning system, which is primarily inspired from MDP to implement a deep reinforcement learning model called as Reinforced Decision Process (RDP). The Reinforced Decision Process (RDP) is an adaptable model, which is inherited from the Markov Decision Process (MDP) that trains the model based on reinforcement learning. It can be very well applied into a progressively updating traffic light convergence, which requires more stages to process, for example, an intermittent crossing point with five or six different ways to implement the same. In such a situation, if there are more stages at a convergence in the intersection point of lane, it can make use of the MDP model as an enhanced measurement reverence. In case of rush hour gridlock, the stages of MDP gets upgraded and the traffic light signal steadily changes in accordance with the agreed measures of the process. The Green and Red signs are predominantly used, but the Yellow sign is required only to ensure safety and security between two neighboring stages, which permits the running vehicles to stop before the signals become red. The Average Waiting Time (AWT) of vehicles is considered as the utmost factor for intelligent traffic monitoring system, which tries to limit the negative impact on the transport economy and enables smooth and easy traversing of vehicles.

The system for smart traffic control management with efficient performance optimization is proposed. Fig.1 shows the outline of the proposed architecture for adaptive traffic control system framework. Two modes of operation are considered in this system namely; the normal mode of operation and the emergency mode of operation. This tries to adapt to the dynamically changing traffic needs and evenly distributes the traffic by load balancing the vehicles on heavily crowded lanes. The intersection of lanes distributed on an X-Y plane is analyzed by unfolding the split up of traffic densities (W_{hy}) across various lanes in the given X-Y plane of representation, as shown in Fig.1.

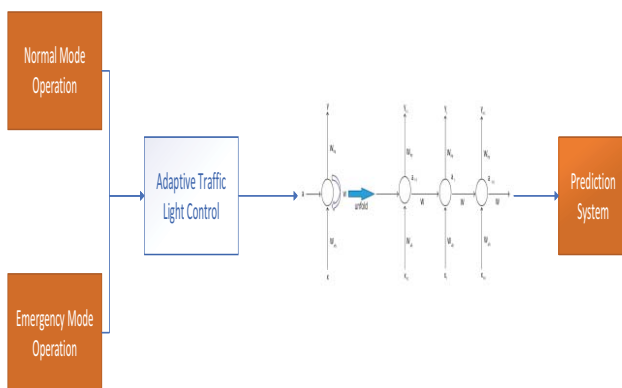


Fig. 1: Layout architecture of Adaptive Traffic Control System

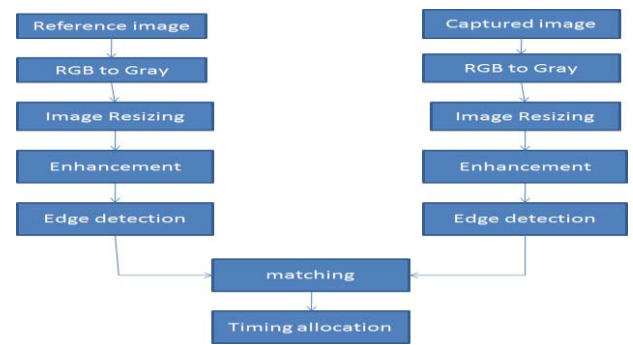


Fig. 2: Proposed working process flow for image capture

The proposed working flow chart for the image capture of traffic detection across the lanes and intersection is represented in Fig. 2. The reference image and captured image are converted into gray scale values and resized with enhancements. The edge detection is primarily done to compare the matching between both the images. Based on this, time is allocated for traffic light signals to efficiently manage and direct the traffic regulation across the lanes.

Fig. 3 depicts the flow process of Raspberry Pi operation using which image is captured, compared and match is sent to cloud with the processed timing of the traffic signal in operation.

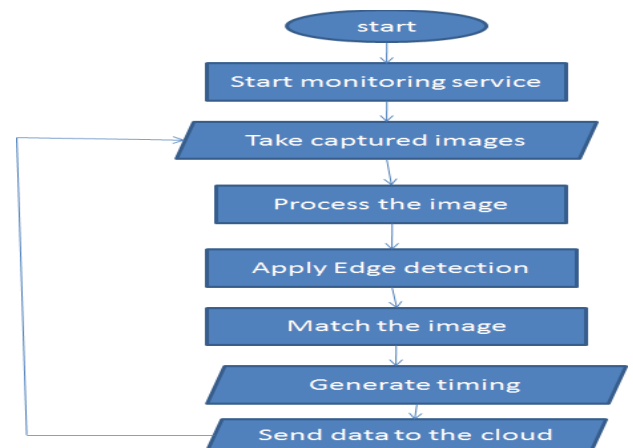


Fig 3: Process flow for Raspberry Pi Operation

The Reinforced Decision Process (RDP) algorithm as procedure is discussed as given below:

RDP Algorithm

- Step 1: Start the process to capture the image of vehicles in traffic lanes and intersection with R_i , C_i as reference image and captured image.
- Step 2: Assign the traffic lights as $G1-R2$ as Green signal for R_i and red signal for C_i .
- Step 3: Assign the traffic lights as $R1-G2$ as Red signal for C_i and green signal for R_i .

- Step 4:** Assign the traffic lights as $R1-R2$ as Red signal for both R_i and C_i .
- Step 5:** Assign the traffic lights as $maxG1minR1$ as Green signal for "Empty lane"
- Step 6:** Assign the traffic lights as $maxR1minG1$ as Green signal for "Crowded lane"
- Step 7:** Assign G_i and R_i to allocate the timing for the Green and Red signals respectively.
- Step 8:** Iterate the steps continuously from 1 to 7 until required output is reached.
- Step 9:** Stop the process.

IV. EXPERIMENTAL RESULTS

The simulation is performed using Python, Numpy, Sci-learn and Eclipse IDE as the backend environment and Web technology and Bootstrap as front end environment to simulate the traffic in real-time scenarios. The experimental results are tested by adapting deep reinforcement learning methodology, which embeds the two-fold Q-learning system.

The experimental outcomes show that the proposed Simulation of Urban Mobility (SUMO) in a vehicular system using Reinforced Decision Process (RDP) algorithms, which makes use of reinforcement learning, based on Markov Decision Process (MDP) algorithm. This ensures the reduction of average waiting time of vehicles and tends to achieve high throughput in comparison to the existing conventional fixed-time control algorithms.

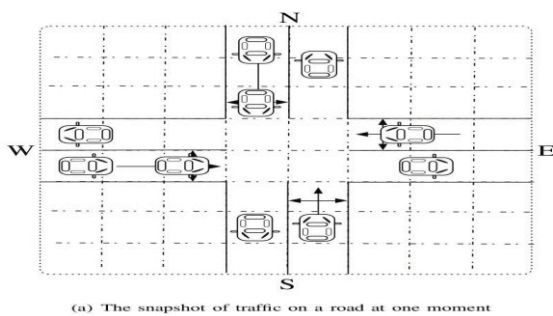


Fig. 4: Sample traffic scenario

Fig. 4 depicts the sample real-time traffic scenario in which the intersection junction of the lanes is considered with the traffic light signalling the transportation of vehicles using dynamic and self-adaptive traffic regulation. Fig.5 shows the intersection scenario tested in the simulation environment for urban mobility of vehicles in adaptive traffic set-up.

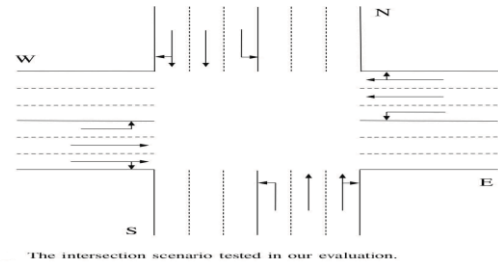


Fig. 5: The intersection scenario with A-ring road type, tested in the simulation environment

Fig.6 illustrates the actual depiction of traffic in real-time environment with deep Reinforcement learning to properly regulate the traffic without any violation, by minimizing the AWT of vehicles and enhancing the overall dynamicity of traffic lanes. The image of vehicle is captured and processed with the aid of image processing and efficiently normalize the traffic in crowded lanes with the traffic signal light ensuring the right movement of vehicles in the lanes of intersection to minimize the grid-lock. Fig. 7 shows a sample A-Ring Road, which is embedded in our model to reduce congestion due to irregular traffic conditions.

Fig.8 shows the graph representing a comparative analysis between the existing Markov Decision Process (MDP) algorithm and the proposed Reinforced Decision Process (RDP). The average waiting time of vehicles in the traffic flow is represented with the time taken and number of vehicles considered.

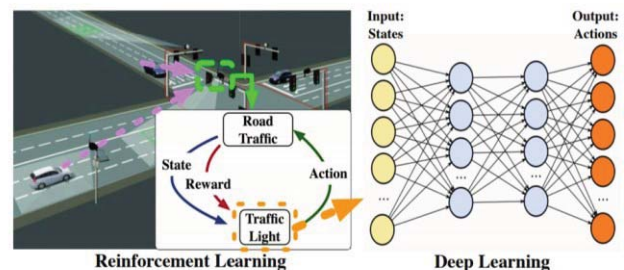


Fig. 6: The proposed real-time scenario tested using deep Reinforcement Learning



Fig. 7: Sample A-Ring Road model to reduce congestion

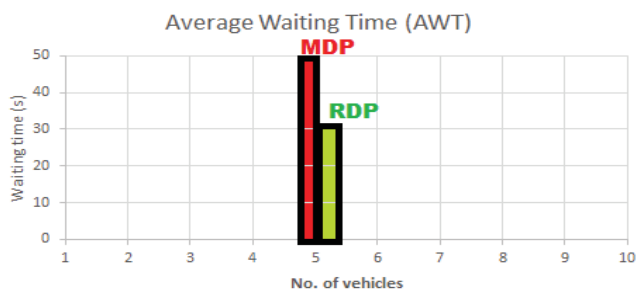


Fig. 8: Graph depicting the Average Waiting Time (AWT) of traffic flow in MDP algorithm and proposed RDP algorithm with time taken and number of vehicles ($n=5$) in the graph

The emergency type vehicles like ambulance, fire and police are given priority by capturing the images and identifying the free routes in lanes to divert the vehicles for ease of transportation.

The simulation results thus obtained show that the proposed system has significantly achieved a self-adaptive and dynamic traffic management system that minimizes the Average Waiting Time (AWT) of vehicles and thereby prioritizes the emergency services to avoid mishaps and congestion.

V. CONCLUSION AND FUTURE WORK

Thus, the proposed system is an improvement of traffic guideline and control in the urban areas to upgrade to smart city. It is primarily reliant on utilizing smart frameworks for checking, controlling and properly regulating the traffic. A productive versatile traffic guideline framework is executed by detecting the traffic thickness or traffic densities on all the streets at a traffic light intersection. It is exceptionally proficient in the event of repetitive aggregation of traffic. The versatile idea of this framework ends up being exceptionally proficient in managing the variable rush hour gridlock densities on streets, by reducing the queue length. Additionally, by giving traffic thickness data to drivers the traffic can be very much overseen. A directed traffic stays away from blockages

and forestalls mishaps. The experimental outcomes show that the proposed Simulation of Urban Mobility (SUMO) in a vehicular system using Reinforced Decision Process (RDP) algorithms, which makes use of reinforcement learning, based on Markov Decision Process (MDP) algorithm. This ensures the reduction of average waiting time of vehicles and tends to achieve high throughput in comparison to the existing conventional fixed-time control algorithms. This in turn predicts the traffic by efficiently minimizing the Average Waiting Time (AWT) of vehicles and thereby prioritizes the emergency services to reach on time by avoiding mishaps and unpredictable traffic congestion.

As part of future work, the system can be enhanced to model the traffic management framework for isolated intersections and semi-urban lanes to autonomously monitor the traffic and self-adapt to dynamic traffic changes in forbidden, anonymous and isolated intersections.

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